

Physiological impacts of salinity treatments on *Carcinus maenas* and *Hemigrapsus oregonensis*

Abstract

Salinity is a key environmental factor in the intertidal environment, impacting the physiological functioning of marine organisms. The invasive *Carcinus maenas* and native *Hemigrapsus oregonensis* both exhibit wide salinity tolerances and geographic overlap. The broad environmental tolerance of these species is known to contribute to their large ranges, however, the physiological tolerance to salinity as an environmental factor, remains less understood. In this study we evaluate physiological responses of both *Carcinus maenas* and *Hemigrapsus oregonensis* in a low, elevated, and ambient salinity condition. We found salinity to have little effect on physiological responses of righting time, respiration, and osmolality. We also found *Carcinus maenas* and *Hemigrapsus oregonensis* to have similar resilience to shifts in salinity. Our results demonstrate that salinity alone does not drive the physiological responses underlying distribution and competition of these species. This highlights the need for further work examining potential interactions of environmental factors and their impacts on physiological function. Understanding the physiology of these crabs can help inform management plans for coastal habitats and the spread of invasive species.

Introduction

Intertidal habitats are incredibly dynamic environments, commonly inhabited by crab species, which utilize a variety of physiological mechanisms to survive fluctuations in temperature, water availability, dissolved oxygen, and salinity. The invasion of European green crab, *Carcinus maenas*, to the west coast of North America has been rapid and successful, largely due to its ability to survive in disparate environmental conditions. The resilience of physiological functioning of *C. maenas* across temperature impacts is well documented (Kelley et al. 2014, Mc Gaw et al. 2012, Rivera et al. 2011, Rivers et al. 2024). Less is known about the salinity tolerance of *C. maenas*, although emerging research identifies the species to be very euryhaline, tolerating salinities ranging between <5 ppt and >50 ppt (Dalpont et al. 2024). Comparatively, the native hairy shore crab, *Hemigrapsus oregonensis*, has a salinity range of <4 and >40 ppt (Falconer et al. 2019), making these species fierce intertidal competitors. Salinity is an important environmental factor impacting the physiology, and subsequent distribution, behavior and life history of many aquatic ectotherms. Exposure to salinity extremes are typically occurring on acute temporal periods in the intertidal, rather than long term exposures. By discerning impacts at the physiological level over longer periods, we can further project habitat viable for potential *C. maenas* invasion, along with strengthen our understanding of competition between these sympatric crab species. The aim of this study was to investigate the physiological impacts of *C. maenas* and *H. oregonensis*, enduring elevated (45 ppt), low (15 ppt), and environmental (35 ppt) salinity treatments. Crabs were held in these conditions over the course of three weeks, with physiological condition evaluated by hemolymph osmolality, righting time, and respiration intermittently. We found there to be minimal decrease in physiological performance in stressful salinity conditions. Righting time and respiration showed no relationship to salinity, and osmolality indicated potential physiological impairment at 45 ppt for *H. oregonensis*.

Methods

Experimental design

Using 3 *C. maenas* and 12 *H. oregonensis* we evaluated the hemolymph osmolality, righting time, and respiration across three weeks, and three salinity treatments at ambient temperature. One *C.*

maenas and four *H. oregonensis* were placed in each salinity treatment, 15 ppt (low), 35 ppt (environmental control), and 45 ppt (elevated). Prior to experimentation, each crab was massed, the righting time was recorded, along with respiration and a baseline hemolymph osmolality. Throughout the experiment, mass, righting time, and respiration rate were measured each week for every crab, and hemolymph osmolality was measured at the very end of the experiment. For **righting time**, crabs were flipped onto their carapace and held with a finger for one second. Once the finger was removed, a timer started and was stopped once the crab “righted”. Righting time was always collected first as it was inferred to be the least stressful of the methods. **Respiration** was measured using a resazurin assay. Individual crabs were placed into separate chambers with enough resazurin to submerge them completely, along with a blank chamber containing no crab. For *H. oregonensis*, 13 mL was required to submerge the crabs and for *C. maenas*, 75 mL was required. Aliquots of 200 μ L were collected at 0, 30, and 60 minute time intervals and transferred to a 96 well plate. Fluorescence was measured using a spectrophotometer plate to quantify resorufin production. After the 60 minute duration, the crabs were rinsed and placed back into their respective salinity conditions. **Hemolymph** was extracted from crabs by inserting a small needle into the proximal most joint of the crab leg. The extracted hemolymph was then analyzed using an osmometer where 10 μ L of hemolymph is pipetted onto a paper round in the tray. An additional two *H. oregonensis* were used for pre-experimental hemolymph osmolality data, to provide a comparison for post-experimental values. These crabs were not re-entered into the experiment after, as crabs can struggle to survive following hemolymph extractions, especially in stressful environmental conditions. A pre-experimental hemolymph osmolality was not gathered for *C. maenas* due to limited supply of these crabs. **Analysis** was completed using R version 4.5.2. When analyzing hemolymph osmolality, A two-way ANOVA test was performed to assess the statistical difference between pairwise species-salinity tests, along with two-sided t-tests. It is important to note a portion of crabs died one day in the study period due to an ammonia spike. This limited our sample size and reduced the amount of data for our analysis, as crabs were not individually labeled. A European green crab also molted, which likely affected respiration rates due to the increased physiological and metabolic demands.

Results

Salinity condition showed no relationship to physiological indicators of righting time, respiration, and hemolymph osmolality.

No statistical evidence supported righting time to increase with change over the course of three weeks in lowered or elevated salinity, for both species of crabs, as was hypothesized. In low salinity (15 ppt), righting time for *C. maenas* during week one (6.58s) showed a decrease to week three (0.98s) (Fig 1a). In elevated salinity (45 ppt), a drop in righting time also occurred, from week one (6.58s) to week three (2.46s). An increase in righting time did occur at ambient salinity (35 ppt) from week one (3.75s) to week three (30.1s) however this followed a molt. Similarly, *H. oregonensis* did not demonstrate an increase in righting time. At 15ppt, week one righting time (1.77s $\pm$ 0.33) decreased to week three (0.74s $\pm$ 0.49) (Fig 1b). At 45 ppt, there was also a decrease in righting time from week one (2.96s $\pm$ 2.56) to week three (1.95s $\pm$ 1.78). At 35ppt a slight increase was observed from week one (1.85s $\pm$ 0.64) to week three (2.23s $\pm$ 1.64).

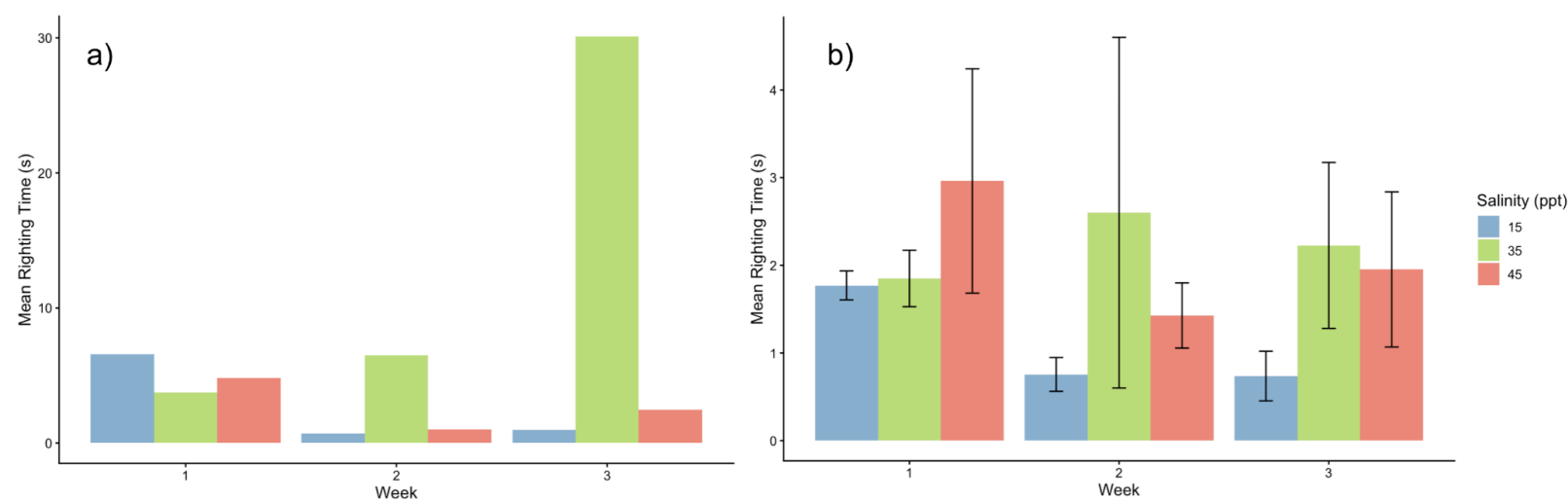


Figure 1 a) Righting time of *C. maenas* (EGC) across salinity conditions (15, 35, 45 ppt) over the three week experimental period. Note, due to supply availability replicates were not present for EGC and therefore no error bars are present. **b)** Mean righting time of *H. oregonensis* (HO) across salinity conditions (15, 35, 45 ppt) over the three week experimental period. Standard error bars are indicated in black.

Resazurin treatments showed a decline in respiration low and elevated salinities, contrary to the hypothesized increase in respiration rate. In low salinity (15 ppt), a decline in *C. maenas* (EGC) marginal mean of respiration fluorescence from week one (2857) to week three (2333) (Fig 2). The same was observed in elevated salinity conditions (45 ppt) where respiration in week one (2982) declined by week three (2168). For *H. oregonensis* at low salinity, a decline in marginal mean in respiration from week one (5953 964) to week three (4739 480) was observed. Similarly, at elevated salinity, a decline from week one (4395 964) to week three (3858 480) was observed.

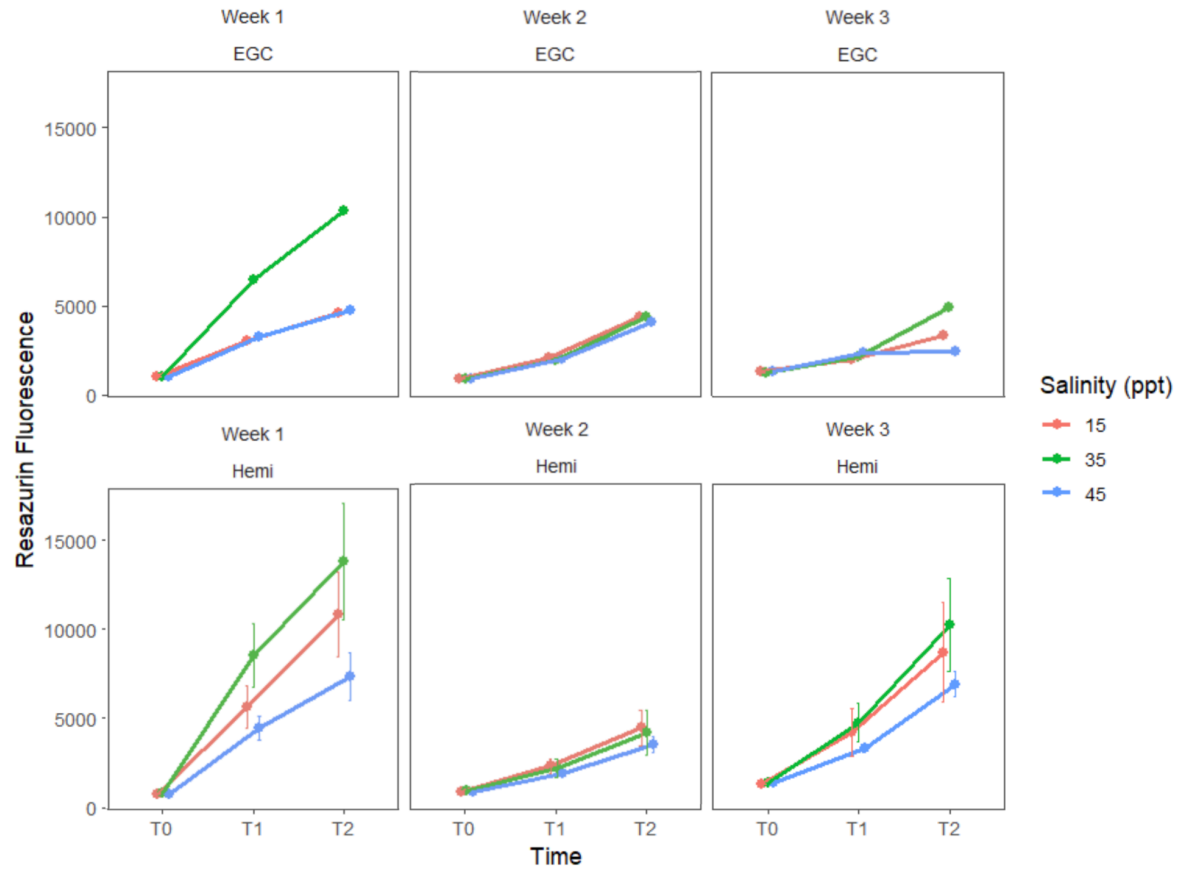


Figure 2 Crabs were submerged in fluorescence of resazurin, respiring over time. Fluorescence is a proxy for oxygen respiration. Each time step represents 30 minute intervals with T0 being the baseline.

We ran a two-way ANOVA test to evaluate statistical differences between pairwise species-salinity tests from which we observed no statistical significance in species and salinity. We further tested each salinity group with *H. oregonensis* independently against the acquired baseline hemolymph osmolality (957.5 mm/kg). In doing so, we found crabs in the 45ppt treatment to have an osmolality (1311 ± 105.5 mm/kg) significantly greater than the baseline (Fig 3). No significant relationship was found in other salinity treatments, which also increased from the baseline osmolality. A baseline was not possible to obtain for *C. maenas* due to limited supply, however osmolality was greatest in the ambient salinity condition (35 ppt).

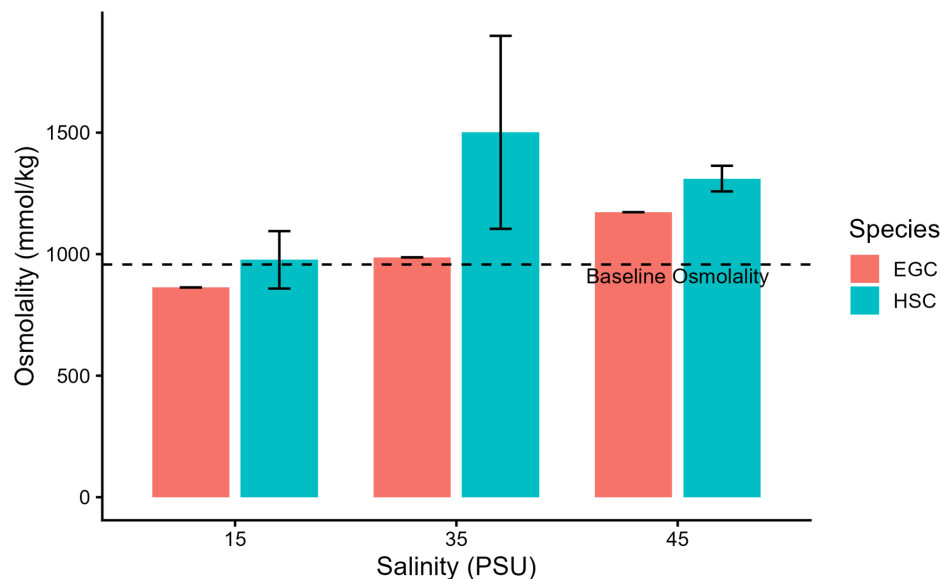


Figure 3 Osmolality (mmol/kg) of hemolymph after 3 weeks of sustained salinity treatments indicated on x-axis (15, 35, 45 ppt). Red bars indicate *C. maenas* (EGC) and display one crab per treatment and therefore no standard error. Blue bars indicate *H. oregonensis* (HSC) and display 3 crabs for 15 and 35 ppt, and 4 crabs for 45 ppt with standard error bars given those populations. Baseline osmolality measurement (dashed line) taken from 2 averaged, non-experimental *H. oregonensis* before the start of the experiment.

Discussion

Overall, salinity did not demonstrate a physiological impact on both *C. maenas* and *H. oregonensis*. Lowered (15 ppt) salinity did not exhibit strong effects on righting time, respiration, and osmolality. Elevated (45 ppt) salinity also did not exhibit strong effects on righting time and respiration, however osmolality regulation was impacted, indicating potential heightened physiological stress.

Righting time did not show the hypothesized relationship with salinity, in either species indicating no physiological stress. Righting time provides insight to physiological performance since it is a metric of bodily function and performance, therefore a shorter righting time is reflective of greater physiological condition (Neil et al. 1993). Righting time did dramatically increase for *C. maenas* in 35ppt at week 3, standard of the Puget Sound region. This was not a condition meant to induce physiological responses, rather to control for ambient salinity. This spike in righting time is likely due to a recent molt event in this crab, where performance of musculature is impacted by the redirection of energy towards rebuilding the exoskeleton (Chang et al. 1995).

Respiration does not indicate physiological stress induced by low or elevated salinity. Decreases in the marginal means across each week for both *C. maenas* and *H. oregonensis* are reflective of a lower respiration. During physiological stress, metabolic demand is expected to increase during osmotic stress (Boeuf et al. 2001). However, fluorescence measurements did decline.

Osmolality changes indicate challenges in regulating to meet environmental salinity shifts. In the case of osmolality, no significance in the ANOVA indicates that salinity did not have an impact on osmolality regulation. When reared at 45 ppt in *H. oregonensis* did demonstrate a decrease in osmolality regulation, potentially due to excess stress. However, since so few crabs were present in each treatment and in the baseline measurements, it is difficult to determine if this is reflective of larger biological implications.

The lack of large changes in these metrics suggest that salinity did not have the anticipated effect on crab physiologically observed previously when evaluating salinity or other environmental stressors, namely temperature (Cuculescu et al. 1998). Invasive *C. maenas* and native *H. oregonensis* showed largely similar responses to environmental stressors. Across the physiological metrics and salinity variables, neither species demonstrated a physiological response that would suggest outcompeting the other. As management plans for invasive *C. maenas* emerge, understanding the responses to salinity extremes by both of the species is incredibly valuable. The wide salinity tolerance in *C. maenas* does explain invasion success across coastal habitats. While this work does not suggest *C. maenas* to have a physiological advantage over *H. oregonensis*, salinity is only one environmental factor considered. Within the intertidal environment, salinity conditions exist at extremes beyond what was tested in this study. Moreover, this highlights the potential for other environmental factors to be driving competition between these species. We identify the potential future work to investigate the impact of other abiotic factors in marine environments and their interactions with salinity, on the physiology of *C. maenas* and *H. oregonensis*.

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